

Answers learning task 6

2026 – 2027

Exercise 4.51: Assume that Z has a standard normal distribution. Determine the value for z that solves each of the following:

- (a) $P(Z < z) = 0.9$
- (b) $P(Z < z) = 0.5$
- (c) $P(Z > z) = 0.1$
- (d) $P(Z > z) = 0.9$
- (e) $P(-1.28 < Z < z)$

Answer

- (a) 1.2816
- (b) 0
- (c) 1.2816
- (d) -1.2816
- (e) 1.2831

Exercise 4.55: Assume that X is normally distributed with a mean of 5 and a standard deviation of 4. Determine the following:

- (a) $P(X < 11)$
- (b) $P(X > 2)$
- (c) $P(3 < X < 7)$
- (d) $P(2 < X < 8)$
- (e) $P(-2 < X < 9)$

Answer

- (a) 0.9332
- (b) 0.7734
- (c) 0.3829
- (d) 0.8013
- (e) 0.5467

Exercise 4.56: Assume that X is normally distributed with a mean of 5 and a standard deviation of 4. Determine the value for x that solves each of the following:

- (a) $P(X > x) = 0.5$
- (b) $P(X > x) = 0.95$
- (c) $P(x < X < 7) = 0.2$
- (d) $P(3 < X < x) = 0.95$
- (e) $P(-x < X - 5 < x) = 0.99$

Answer

- (a) 5
- (b) -1.5794
- (c) 4.9144
- (d) $\emptyset$
- (e) 10.3033

Exercise 4.63: The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean of x cc and a standard deviation of 5 cc.

- (a) At what value should the mean be set so that 99.9% of all cans exceed 365 cc?
- (b) At what value should the mean be set so that 99.9% of all cans exceed 365 cc if the

standard deviation can be reduced to 2 cc?

Answer

- (a) 380.4512
- (b) 371.1805

Exercise 4.69: The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours.

- (a) What is the probability that a laser fails before 6000 hours?
- (b) What is the life in hours that 95 % of the lasers exceed?
- (c) If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours?

Answer

- (a) 0.0478
- (b) 6013
- (c) 0.125

Exercise 4.91: Suppose X has an exponential distribution with mean equal to 10. Determine the following:

- (a) $P(X > 10)$
- (b) $P(X > 20)$
- (c) $P(X < 30)$
- (d) Find the value of x such that $P(X < x) = 0.90$.

Answer

- (a) 0.3679
- (b) 0.1353
- (c) 0.9502
- (d) 23.0259

Exercise 4.103: The lifetime of a mechanical assembly in a vibration test is exponentially distributed with a mean of 400 hours.

- (a) What is the probability that an assembly on test fails in less than 200 hours?
- (b) What is the probability that an assembly operates for more than 500 hours before failure?
- (c) If an assembly has been on test for 400 hours without a failure, what is the probability of a failure in the next 100 hours?
- (d) If 10 assemblies are tested, what is the probability that at least one fails in less than 100 hours? Assume that the assemblies fail independently.
- (e) If 10 assemblies are tested, what is the probability that all have failed by 800 hours? Assume that the assemblies fail independently.

Answer

- (a) 0.3935
- (b) 0.2865

- (c) 0.2212
- (d) 0.9179
- (e) 0.2336

Exercise 4.165: Suppose that $f(x) = 0.5x - 1$ for $1 < x < 4$. Determine the following:

- (a) $P(X < 2.5)$
- (b) $P(X > 3)$
- (c) $P(2.5 < X < 3.5)$
- (d) Determine the cumulative distribution function of the random variable.
- (e) Determine the mean and variance of the random variable.

Answer

- (a) 0.0625
- (b) 0.75
- (c) 0.5
- (d) $F(x) = \begin{cases} 0, & x \leq 2 \\ 0.25x^2 - x + 1, & 2 < x < 4 \\ 1, & x \geq 4 \end{cases}$
- (e) $E(X) = 3.3333, \quad V(X) = 0.2222$